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APPLIED RESEARCH

Transformer-Based OCR System for Handwritten Signature Verification in Lawmaking Petitions: A Proof of Concept

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ABSTRACT Handwritten signature verification (HSV) is essential for validating lawmaking petitions, where each signatory provides a handwritten signature and a personal identifier (ID). Manual verification is slow and error-prone due to large petition volumes and handwriting variability. This paper presents a methodology for automated HSV: an end-to-end transformer-based OCR pipeline built on TrOCR, a state-of-the-art model for text recognition. While TrOCR focuses on cropped text segments, the proposed methodology extends it to full-document, integrated processing. It performs document structure detection, ID recognition, handwritten signature detection, and ID–signature pairing within a unified framework. As part of the methodology, we developed a synthetic handwriting generation method to create training data, addressing data scarcity and protecting privacy without using real personal information. The system was developed in collaboration with the Parliament of the Canary Islands, ensuring institutional alignment and practical applicability. Evaluation considered post-processing, model fine-tuning, and dataset scale. The pipeline achieved 98.9% character recognition and 90.5% complete ID recognition on synthetic data, and 84.3% character recognition and 80.0% ID recognition on real petition images. In comparison, a convolutional recurrent neural network (CRNN) baseline achieved 81.2% character recognition and 12.6% ID recognition on synthetic data, and 66.2% and 7.4% on real petitions. Post-processing improved recognition for non-fine-tuned models, while fine-tuning on larger datasets was crucial for high accuracy and generalization. In conclusion, this methodology demonstrates that extending TrOCR into a full-document, transformer-based pipeline provides a scalable and privacy-preserving solution for HSV in legislative processes, outperforming conventional neural networks and reducing manual verification workload.

INDEX TERMS OCR, transformers, handwritten ID recognition, synthetic datasets, lawmaking, proof of concept.

I. INTRODUCTION

Lawmaking petitions allow citizens to request new legislation, amendments to existing laws, or advocate for policy reforms, ensuring that the public's voice is represented in legislative processes and decision-making. These petitions address critical domains such as economic development, healthcare, education, transportation systems, and sustainable

urban growth, aligning governmental actions with societal priorities [1], [2].

The petition process in parliamentary settings generally involves four stages: submission, admissibility assessment, consideration, and closure. Petitions may be either handwritten or electronic, and they can be submitted in physical or digital form, depending on the legislative framework of the respective jurisdiction. Despite the growing adoption of electronic petition systems, handwritten lawmaking petitions submitted in physical form using standardized templates

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(hereafter written petitions) remain a central element of the petition process in many European parliaments and continue to serve as the primary method in various regions world-wide [3].

For a written petition to be deemed admissible, it must contain a valid handwritten signature from each signatory and meet the minimum number of signatories required by law, typically ranging from tens of thousands to several hundred thousand, depending on jurisdictional and demographic factors. A valid handwritten signature includes both a personal identifier (hereafter ID), which is a unique alphanumeric code assigned to each individual, and the personal signature. Figure 1 presents an example of a simulated written petition containing three human-performed handwritten signatures that do not include real personal data, as might be submitted to the Parliament of the Canary Islands, Spain. In the example, each ID associated with each signatory comprises nine characters: eight digits followed by one letter, consistent with the format defined by the Spanish Home Office [5].

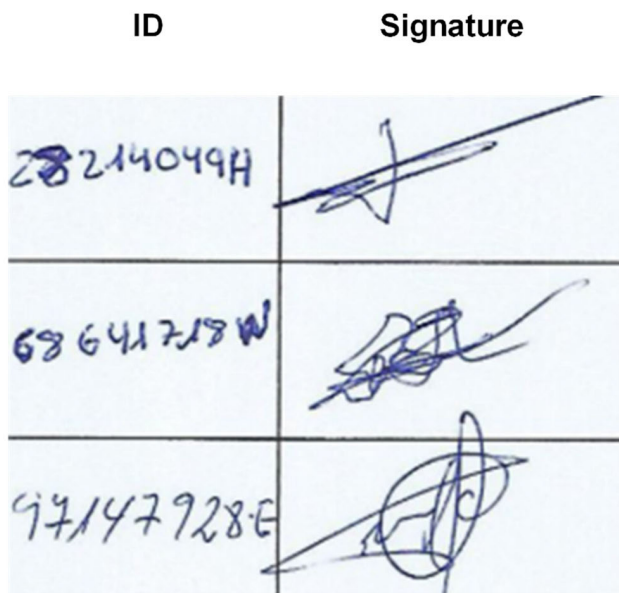


FIGURE 1. Cropped section of a simulated written petition image illustrating three human-performed handwritten signatures, each comprising an ID and signature that identify the respective signatories endorsing the simulated petition.

Handwritten signature verification (HSV) is a critical process for determining the admissibility of a written petition. It entails the detection and recognition of each signatory's ID, the validation of its authenticity, and the confirmation of a corresponding signature. Traditionally, this process has been carried out manually, making it labor-intensive, time-consuming, and prone to human error, given the high volume of signatories and the variability in handwriting styles. Furthermore, verification inaccuracies, such as the failure to identify a valid ID, can lead to the rejection of a petition if the legally required threshold of valid signatories is not met.

To address these issues, approaches like Optical Character Recognition (OCR), a challenging research topic in computer vision (CV) and natural language processing (NLP), provide a viable solution for converting scanned handwritten documents into machine-readable formats [6] and automating the recognition of handwritten IDs while confirming the presence of the corresponding signature. Figure 2 presents the architecture of the OCR pipeline designed for the HSV task applied to written petitions. These petitions follow a form-based format, and the pipeline consists of four key modules:

- **Scanning:** Physical documents containing the written petitions from signatories are digitized into images.
- **Text detection:** The image is segmented into meaningful components. Each component is identified by a bounding box, one enclosing the signatory's handwritten ID and another enclosing the corresponding signature.
- **Text recognition:** Each ID is recognized by the recognizer component, while each signature is detected by the signature detector.
- **Results:** The validity of each recognized ID is verified, ensuring that it is paired with a signature. A report is generated based on the results using predefined metrics.

Traditional OCR pipelines primarily rely on machine learning (ML) approaches, particularly Convolutional Neural Networks (CNNs) [7] for handwritten text detection and recognition tasks. While CNNs have proven effective in image classification, object detection, and printed text recognition, they face limitations when applied to handwritten text. These limitations include the need for a separate language model, such as Recurrent Neural Networks (RNNs) [8], to convert visual features into text sequences. This additional step can introduce errors due to misalignment between the CNN output and the language model predictions, making the system challenging to scale for handwriting styles [9]. Consequently, such models are unsuitable for the HSV problem addressed in this paper, where verification mistakes can invalidate valid petitions.

The emergence of transformer-based models offers a new approach to be explored. Their ability to capture long-range dependencies through self-attention mechanisms makes them particularly well-suited for sequence-based tasks like text detection and handwriting recognition. Despite recent advances in the application of transformers to these tasks, several challenges persist in the context addressed by this paper. These include: investigate their adaptability to previously unseen data, substantial data requirements, as their effective performance depends on large amounts of annotated data, that in this context are scarce and costly to obtain; difficulties in fine-tuning the model for the specific domain, which requires retraining the model on labeled datasets; the need to comply with privacy laws such as the General Data Protection Regulation (GDPR) [4], since datasets contain sensitive personal information like IDs and signatures; and high computational demands, which can hinder deployment.

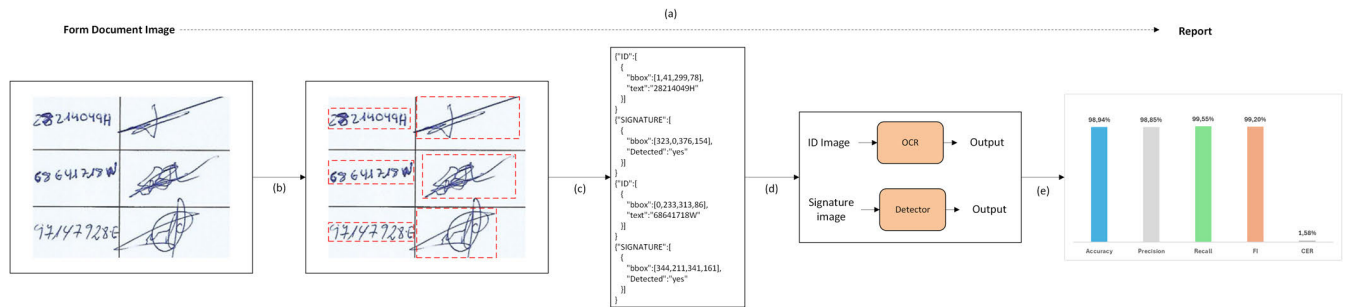


FIGURE 2. OCR Pipeline for HSV applied to Form-based Documents. (a) The objective is to identify and validate each ID, confirm that it is accompanied by a signature, and generate a report based on predefined metrics. (b) The pipeline begins with converting physical documents into digital images. (c) Text detection follows, locating the IDs and signatures of each signatory. (d) Each detected ID is passed to a recognizer, while each detected signature is sent to a signature detector to confirm its presence alongside the ID. (e) The validity of each recognized ID is verified, and a report summarizing the results is generated.

in resource-constrained environments such as parliamentary settings.

This paper introduces a proof of concept (PoC) OCR system based on a transformer-centric solution to address the challenges of HSV applied to written petitions. The proposed system not only automates the HSV process but also tackles data scarcity by developing privacy-compliant synthetic datasets that simulate handwriting styles. Developed and evaluated in collaboration with the Canary Islands Parliament, the system complies with institutional legal and technical constraints and demonstrates significant improvements in accuracy compared to traditional CRNN-based frameworks. The main contributions are summarized as follows:

- Transformer-based approach: Introducing a method for HSV applied to written petitions using exclusively transformer-based models, which enhances accuracy compared to CRNN-based frameworks.
- Privacy-compliant synthetic datasets: Developing synthetic datasets of varying sizes that simulate handwriting styles while adhering to GDPR. This approach addresses data scarcity and evaluates the effectiveness of model training.
- Viability of transformer-based models for real-world problems: Investigating the transferability and adaptability of transformer models for detecting and recognizing IDs applied to written petitions.
- Comprehensive Evaluation: Assessing performance using a verification, validation, and testing (VVT) framework with dual metrics: (1) ML-based ID-level accuracy for evaluating the system's ability to correctly recognize IDs as whole entities and (2) NLP-based character-level accuracy for fine-grained text recognition errors.
- Parliamentary application: Implementing and evaluating the system within a real-world parliamentary setting, where computational resources are limited, demonstrating its practical feasibility and effectiveness for the automation of a manual process.

The paper is organized as follows: Section II discusses related work; Section III describes the methodology; Section IV details the specific methods and materials used, with a focus on a transformer-centric approach; Section V presents the results; Section VI discusses the findings; and Section VII provides concluding remarks.

II. RELATED WORK

This section reviews prior research relevant to the proposed transformer-based OCR system for handwritten signature verification in lawmaking petitions. The discussion is structured as follows: first, we examine OCR approaches; second, we review methods for signature detection and verification in legal contexts; and finally, we present synthetic handwriting generation techniques, emphasizing approaches most relevant to the objectives of our work.

A. OPTICAL CHARACTER RECOGNITION

Traditional OCR systems are typically divided into two main components: text detection and text recognition. The detection stage localizes text regions within an image and is generally formulated as an object detection problem. The recognition module then transcribes the visual features of each localized region into machine-readable text. This is commonly modeled as an encoder-decoder process, where the encoder extracts a compact representation of the visual information, and the decoder sequentially generates the corresponding textual sequence.

Recent deep learning-based OCR approaches can be broadly categorized into two groups: (1) CRNN-based models, which combine convolutional and recurrent layers with transcription modules, and (2) Transformer-based models, which employ self-attention mechanisms for both image encoding and sequence decoding. The following subsections describe how these approaches address text detection and recognition.

1) TEXT DETECTION

The conventional techniques for text detection using CNN-based approaches include Faster R-CNN [10],

Mask R-CNN [11], EAST [12], CRAFT [13], and YOLO [14]. These models are trainable end-to-end, meaning they can directly learn to predict text features from raw image inputs without requiring hand-crafted features or separate steps like manual feature extraction. However, they differ in several aspects: (1) output granularity: some models, such as Faster R-CNN, YOLO, and EAST, predict bounding boxes for detected text regions, localizing the text within the image. Other models, like Mask R-CNN, predict more complex shapes such as masks, which provide pixel-level segmentation of the text regions, offering a more detailed representation; and (2) speed vs. accuracy trade-offs: models like YOLO and EAST are designed to be particularly fast and efficient, making them suitable for real-time applications. They strike a balance between speed and detection accuracy. In contrast, Mask R-CNN and Faster R-CNN are more accurate in text detection but are computationally more expensive, making them better suited for applications where precision is a higher priority than speed.

Text detection in form-based documents involves a preliminary step, such as identifying where tables are located (a.k.a. table detection) and subsequently inferring the table structure from the document by identifying the individual pieces that make up a table, like rows, columns, and cells (a.k.a. table structure recognition) [15]. While earlier approaches were based on general-purpose architectures such as Faster R-CNN [16], current methods often rely on training transformer-based object detection models, such as DETR [17], for both table detection and structure recognition. Currently, these transformer-based models have shown better results compared to Faster R-CNN [18].

2) TEXT RECOGNITION

Early methods treated the problem of recognizing text as a general image classification, focused on CNN end-to-end OCR models [19], [20]. Other approaches utilized a CNN-based encoder for image understanding, capturing spatial hierarchies and local patterns, while an RNN-based decoder was employed for text generation, to handle sequences and predict the next token based on the previously generated ones [38]. While CNNs have proven highly effective in image classification, object detection, and recognizing printed text, and have been foundational in the development of OCR systems, they face different limitations when applied to handwritten text detection and recognition. These limitations include integration issues, ensuring that outputs from CNNs are properly aligned with inputs to RNNs, which adds complexity to the architectural design. Additionally, error propagation is a concern; since RNNs process data sequentially, any error in the CNN output (e.g., misclassifying a character) can propagate through the RNN and lead to incorrect text predictions. CNNs also tend to be inefficient with irregular text, particularly when it is rotated or curved.

Recent progress in deep learning models has led to significant improvements by leveraging transformer-based models,

a deep learning architecture introduced by Vaswani et al. [21] and originally developed for NLP tasks. The key innovation of transformers is the self-attention mechanism, which enables the model to weigh the importance of different words (or tokens) in a sequence, regardless of their distance from each other. This allows the model to attend to different parts of the sequence simultaneously and capture long-range dependencies. This contrasts with earlier sequence models like RNNs and LSTMs, which process tokens in order and struggle to capture long-range dependencies.

Transformer-based models can be trained on massive amounts of text data to learn general representations, resulting in pre-trained language models (PLMs) such as BERT, GPT, and T5. These PLMs can then be fine-tuned with labeled data to perform specific tasks. Transformer-based models have also been adapted for vision tasks, such as Vision Transformers (ViTs), which divide an image into fixed-size patches, treat each patch as a token (similar to words in NLP), and process them using self-attention mechanisms. This approach enables ViTs to model global relationships between different parts of an image more effectively than CNNs, particularly for tasks that require understanding long-range dependencies [22].

Recent work suggests that combining transformer-based models for image and language processing can enhance feature representation by replacing the CNN backbone with image transformers and using language transformers instead of RNNs for more robust textual modeling. TrOCR exemplifies this approach [23], a transformer-based OCR model that employs architectures similar to ViT and DeiT [24] as its encoder and RoBERTa (a pre-trained BERT-style model) [25] as its decoder. The model is initially pre-trained on a large-scale dataset of synthetically generated data for image understanding and language modeling and can then be fine-tuned with labeled datasets for specific tasks. The model focuses solely on the text recognition task, excluding the text detection phase.

While TrOCR has been fine-tuned for various domain-specific applications such as recognizing historical documents in archives [26], interpreting medical prescriptions [9], and processing scanned receipts in financial settings [27], adapting it to the HSV task for lawmaking petitions presents additional challenges. These challenges include handling full-document inputs, detecting and associating handwritten IDs with signatures, and ensuring compliance with privacy regulations. The specific challenges and the methodology used to address them are detailed in the next subsection and further elaborated in Section III.

B. SIGNATURE DETECTION AND VERIFICATION IN LEGAL CONTEXTS

Research on signature detection has primarily focused on specialized domains. For example, Sánchez et al. [39] analyze historical parish registers by segmenting documents into discrete acts, while Cloppet and Stutzmann [40]

investigate segmentation in administrative and archival manuscripts. More recent work has explored transformer-based architectures for legal document analysis. TransDocAnalyser [41], for instance, employs vision transformers for field localization and structured text extraction in semi-structured administrative records, reporting promising results for mixed handwritten and printed text. However, these systems are designed mainly for layout understanding and field recognition rather than addressing the combined tasks of signature presence detection and handwritten identifier recognition required in lawmaking petitions. Consequently, existing approaches are rarely adapted to the specific constraints of legislative petitions, where signatures must be validated jointly with personal identifiers to ensure legitimacy.

In parallel, offline signature verification has long been a central research focus, aimed at distinguishing genuine signatures from forgeries. Convolutional neural networks (CNNs), Siamese architectures, and hybrid CNN–Vision Transformer models have been investigated for this purpose [42], [43], [44]. While these methods demonstrate effectiveness in authenticity verification, they generally fail to address two critical aspects: detecting whether a signature is present in a document and recognizing associated handwritten identifiers such as ID numbers. Díaz et al. [45] provide a perspective analysis of signature verification technologies, highlighting persistent challenges such as dataset scarcity, privacy concerns, and limited generalizability across languages and institutional environments.

To provide a structured overview of these approaches and their respective limitations, Table 1 presents a comparative summary of state-of-the-art OCR and signature verification methods. The table categorizes each method by task and details the primary methodology, reported advantages, identified limitations, and the corresponding research gap in relation to the objectives of this work.

The review of existing methods reveals several limitations in existing OCR and signature verification pipelines. Most approaches address individual tasks such as text detection, text recognition, table structure recognition, or signature verification, and none provide a comprehensive solution capable for handling the full complexity of lawmaking petitions. Methods like TrOCR are effective at recognizing text but only when applied to cropped text instances. DETR and DeeDe SRT focus on table structure recognition but are not designed for handwritten signatures or IDs. Similarly, existing signature verification models, including CNN-based or Siamese networks, do not integrate with the workflow of legal petitions.

This research gap motivates our transformer-based methodology, presented in Section III. We introduce a proof-of-concept that extends TrOCR into a multi-task OCR pipeline for verification of handwritten signatures and IDs in lawmaking petitions. The system handles detection, recognition, and ID-signature pairing within one unified framework,

overcoming the limitations of the TrOCR model that perform recognition only on cropped text instances. Our approach addresses several gaps highlighted in Table 1:

- **Full-document, integrated processing:** Unlike prior methods based on TrOCR model [9], [23], [26], [27], our pipeline simultaneously detects, recognizes, and links handwritten IDs and signatures across multi-entry petition forms, providing an end-to-end solution rather than isolated recognition or verification steps.
- **Transformer-based context modeling:** Captures global dependencies across documents using self-attention, improving the association of IDs with corresponding signatures.
- **Adaptation to legal workflows:** The pipeline is designed specifically for lawmaking petitions, ensuring compliance with procedural and operational requirements, which is not addressed by existing OCR or signature verification approaches.
- **Data-efficient and GDPR-compliant:** The synthetic dataset generation enables training and evaluation without exposing real citizen data, addressing privacy concerns inherent in legislative environments.

These contributions establish our system as the first proof-of-concept, end-to-end transformer-based OCR pipeline for handwritten signature verification in lawmaking petitions. It extends TrOCR by integrating detection, ID-signature pairing, and recognition and verification within a unified system, while ensuring GDPR-compliant training using synthetic datasets. The approach combines methodological innovation and applied research to provide a robust solution for the unique challenges of handwritten signature verification in the legislative domain.

C. SYNTHETIC HANDWRITING GENERATION

The scarcity of large, annotated handwriting datasets is a major obstacle for developing OCR systems for legal documents, particularly when they contain personal identifiers and signatures. Access to real-world data is constrained by privacy regulations such as the GDPR, and available datasets must be diverse and representative to support robust fine-tuning of models like TrOCR, which, although pre-trained on printed and English handwritten text, has not been specifically adapted for handwritten IDs in lawmaking petitions [23].

Synthetic handwriting generation has emerged as a practical solution to overcome dataset limitations. Approaches using Generative Adversarial Networks (GANs) and Recurrent Neural Networks (RNNs) have been applied to produce realistic handwritten text by modeling variability in human writing styles [46], [47]. These methods have proven effective in historical and low-resource scenarios, enabling the generation of diverse handwriting patterns for training recognition systems [48]. However, existing methods primarily focus on generic text synthesis and are not tailored to the

TABLE 1. Comparative summary of state-of-the-art OCR, Table detection, and signature verification methods.

Method	Task	Methodology	Advantages	Limitations	Research Gap
Faster R-CNN [10, 16]	Text Detection	CNN-based object detection	High precision	Computationally expensive	Not designed for handwritten text of legislative petitions
Mask R-CNN [11, 16]	Text Detection / Table Detection	CNN-based detection with mask prediction	Pixel-level segmentation; Table structure extraction	Computationally expensive	Lack adaption to handwritten legal documents
EAST [12]	Text Detection	CNN-based text detector	Fast and efficient for real-time	Bounding-box only; May miss irregular text	Ineffective for irregular handwritten text in petitions
CRAFT [13]	Text Detection	CNN-based character region detector	Accurate at character level;	Requires post-processing for word-level recognition	Not designed for handwritten legal signatures/IDs
YOLO [14]	Text Detection	CNN-based object detection	Fast and suitable for real-time	Lower accuracy compared to Mask/Faster R-CNN	Limited accuracy for handwritten elements; not suited for petition IDs
DETR [17, 18]	Table Detection / Structure recognition	Transformer-based object detection	Better global context modeling	Requires large datasets	Ignores handwritten signatures and ID fields
DeeDe SRT [15,16]	Table Detection / Structure Recognition	Transformer-based table structure recognition	Accurate table structure detection in scanned documents	Computational cost; Performance lower than [17], [18]	Not evaluated for handwritten legal documents
CNN-only [19,20]	Text Recognition	End-to-end image classification	Simple architecture; Effective for printed text	Inefficient for irregular or handwritten text	Fails on handwritten text, signatures, or IDs
CNN + RNN [38]	Text Recognition	CNN encoder + RNN decoder	Captures spatial hierarchies; Handles sequential text generation	Slow training; Error propagation	Not tailored for handwritten IDs/signatures; inefficient due to error propagation
ViT [22]	Text Recognition	Transformer encoder with self-attention; Fine-tuned for downstream tasks	Captures long-range dependencies; pre-trainable	Requires large pre-training; Computationally intensive	Requires specific adaptation for handwritten signatures/IDs
TrOCR [8–9], [23–27]	Text Recognition	Transformer encoder-decoder pre-trained on synthetic data	Fine-tunable for OCR tasks	Only text recognition; separate detection required	Needs extension for full-document processing, integrating signature detection with ID linkage, and requires synthetic datasets for training due to data protection constraints
TransDocAnalyser [41]	Legal Document Analysis	Vision transformers for field localization	Handles semi-structured documents	Focused on layout/text fields, not signatures/IDs	Focuses on layout/text fields, not handwritten IDs or signatures
CNN [42]	Signature Verification	CNN-based feature extraction	Distinguishes genuine signatures from forgeries	Does not detect signature presence; no ID recognition	Handles authenticity only; no ID or detection
Siamese network [43]	Signature Verification	Siamese network for metric learning	Good for one-shot verification	Limited to signature authenticity; ignores context	Limited to signature matching; ignores ID/petition context
Hybrid CNN-ViT [44]	Signature Verification and forgery detection	CNN for local features and ViT for global dependencies	High accuracy; captures local and global patterns	Needs large datasets; tested only on controlled data; no ID integration	Lacks OCR and ID integration for legislative petitions

specific requirements of legal documents, where handwritten identifiers and signatures must be accurately represented and paired.

To address this gap, the present work introduces a synthetic dataset generation methodology designed for handwritten lawmaking petitions. Publicly available handwritten

character datasets (EMNIST) [31], [32], [33] and signature datasets [34] are combined, and data augmentation techniques, including random rotations, Gaussian blurring, and scaling, are applied to produce synthetic datasets for both ID numbers and signatures. These datasets are GDPR-compliant, diverse, and suitable for fine-tuning TrOCR to perform handwritten ID recognition in an integrated, end-to-end pipeline.

This contribution provides a novel resource that supports OCR model training and evaluation under realistic legal conditions. By bridging the gap left by existing synthetic handwriting generation approaches, it enables automated validation workflows for legislative petitions, ensuring that OCR models can be trained effectively while respecting privacy constraints.

III. METHODOLOGY

The proposed OCR system for the HSV task applied to written petitions was designed, developed, and tested collaboratively by researchers from the University of La Laguna (ULL) and the Chief Information Officer (CIO) of the Parliament of the Canary Islands. The methodology follows the established principles of scientific research in information systems design [28] and comprises five stages: (1) problem identification and motivation, (2) definition of solution objectives, (3) system design and development, (4) evaluation, and (5) demonstration.

In the first stage, the problem and its objectives were analyzed within a real-world context, highlighting the need for an efficient and automated system capable of performing HSV with minimal manual intervention.

The system design phase formalized a modular, transformer-based processing pipeline that, unlike prior works applying OCR only to cropped text lines [9], [23], [26], [27], operates end-to-end on scanned petition images by integrating document-level structure detection, modality-specific recognition of identification fields and handwritten signatures, and a cross-modal presence validation step to ensure that each recognized identification field is accompanied by a signature, supporting automation and scalability under parliamentary requirements.

Subsequently, the system was evaluated through a verification, validation, and testing (VVT) framework [29]. The demonstration activity included presenting the system to the CIO of the Parliament of the Canary Islands as part of the testing phase, to assess its practical applicability in an operational environment.

The following subsections provide a detailed account of each stage of the methodology. Subsection A outlines the problem identification and motivation. Subsection B presents the specific problem addressed in this work, discusses the research gap in relation to the fundamental theory of transformer architectures, and summarizes the contributions of the paper. Subsection C describes the solution objectives, while Subsection D explains how the proposed pipeline is designed to be scalable and adaptable across different jurisdictions and

legal settings. Section IV details the proposed OCR pipeline, Section V reports the evaluation results, and Section VI discusses the main findings within the VVT framework.

A. PROBLEM IDENTIFICATION AND MOTIVATION

In the Parliament of the Canary Islands, verifying the voter status of signatories for citizen-proposed bills is a legal requirement currently carried out manually [30]. This process is time-consuming and prone to error and becomes increasingly unsustainable given the high volume of signatories per petition. This challenge is exemplified by the most recent popular initiative bill submitted in March 2023 (reference 10L/PPLP-0036), concerning volcanoes in the Canary Islands, which required the verification of over 22,000 citizens.

Building on this challenge, the next subsection formalizes the problem, identifies research gaps, and highlights our contributions.

B. PROBLEM STATEMENT, RESEARCH GAP, AND CONTRIBUTIONS

The specific problem addressed in this work arises from the need to validate handwritten signatures in conjunction with their corresponding handwritten ID numbers within citizen-driven legislative petitions. In the current manual workflow, signatures are visually inspected, and IDs are cross-checked against electoral registers, a process that is slow, error-prone, and unsustainable given the volume of signatories required by law.

1) PROBLEM STATEMENT

Automating this process requires solving two tightly coupled challenges at the full-document level: (1) detecting the presence and location of handwritten signatures in tabular petition forms, and (2) recognizing and validating handwritten IDs associated with those signatures. Both tasks must be handled at the full-document level, where each page contains multiple entries structured in rows, each consisting of an ID field and a corresponding signature field.

2) RESEARCH GAP

State-of-the-art OCR approaches have advanced either detection or recognition but rarely both within a single pipeline. Transformer-based models such as TrOCR [23] achieve state-of-the-art results in line or cropped-region text recognition but are limited to individual text segments and do not address full-page handwritten documents or signature detection. Recent extensions of TrOCR to full-page OCR in printed receipts [54] demonstrate the feasibility of end-to-end transcription without explicit detection but remain restricted to machine-printed text and cannot process multiple handwritten entries or link signatures to IDs.

Transformer detectors such as DETR [17], [18] excel at object localization but do not perform token-level handwriting recognition. Advanced document understanding models

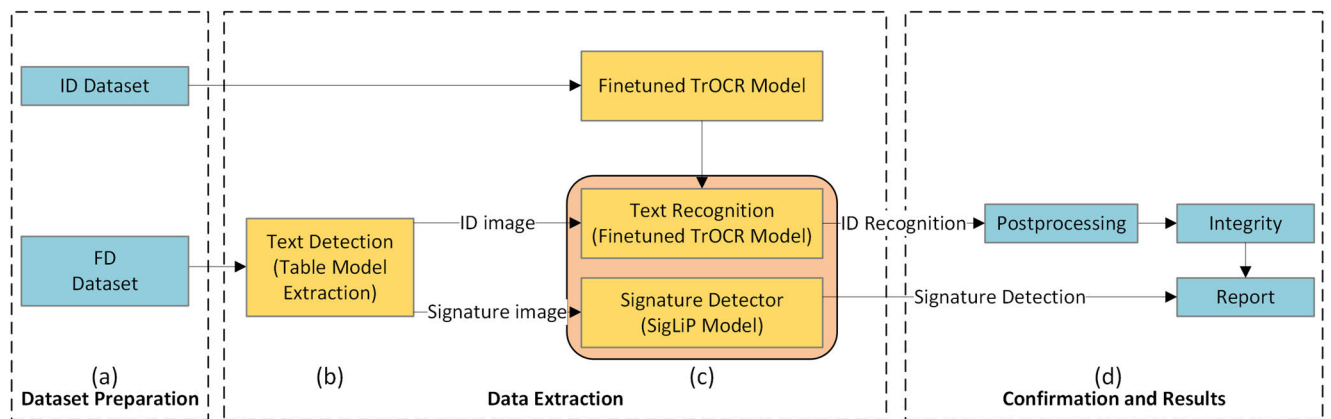


FIGURE 3. OCR Pipeline for HSV Applied to Written Petitions Using Transformer-based Models. (a) The pipeline begins with dataset preparation, including the generation of the ID Dataset for fine-tuning the TrOCR model and the FD Dataset for creating written petition forms used in system evaluation. (b) The text detection submodule locates handwritten IDs and signatures for each signatory using a transformer-based table extraction model. (c) Detected IDs are processed by the fine-tuned TrOCR model to extract the corresponding digital text, while signatures are analyzed using a transformer-based image classification model to confirm their presence alongside the associated IDs. (d) Each recognized ID is validated, and a summary report is generated.

like LayoutLMv3 [49] and DocFormer [50] capture semantic and visual structures yet focus primarily on printed content. Vision-language models for signature verification, e.g., SigLIP [36], provide robust signature detection but are not integrated with OCR pipelines.

From a transformer-theoretical perspective, existing systems underuse self-attention, which allows each token to interact with every other token across the document. This limits contextual modeling across multiple entries and prevents end-to-end integration of detection, recognition, and verification tasks.

3) CONTRIBUTIONS

To address these limitations, we introduce a proof-of-concept end-to-end OCR system for handwritten lawmaking petitions, leveraging transformer architectures at multiple levels:

- Full-document, transformer-based OCR pipeline: Unlike prior TrOCR-based methods limited to cropped lines, our system processes entire petition forms containing multiple handwritten IDs and signatures, enabling simultaneous detection, recognition, and ID–signature pairing.
- Integration of structure recognition and signature detection: We employ a DETR-inspired module for table and structure recognition to localize handwritten ID and signature fields, while a SigLIP-based detector identifies corresponding handwritten signatures. This ensures robust localization of both IDs and signatures within a document’s tabular layout.
- Transformer-based ID recognition. Cropped ID regions are adapted for TrOCR, exploiting self-attention to capture contextual dependencies across diverse handwriting styles.
- Synthetic dataset generation. To address data scarcity and ensure GDPR compliance, we designed a synthetic

dataset generator producing (a) an ID dataset for TrOCR fine-tuning and (b) a form-level dataset for evaluating the integrated pipeline.

- Institutional relevance and novelty. To our knowledge, this is the first system that combines full-document processing, detection, recognition, and ID–signature pairing within a unified transformer-based pipeline tailored to the procedural requirements of parliamentary institutions.

Taken together, these elements indicate that our contributions include the first proof-of-concept for an end-to-end transformer-based OCR system dedicated to handwritten signature verification in lawmaking petitions. Unlike prior approaches, our pipeline extends TrOCR by jointly performing detection, ID–signature association, and recognition–verification tasks within a unified full-document framework. In addition, we introduce a GDPR-compliant synthetic dataset methodology to support training and evaluation under realistic legal conditions. Collectively, these contributions combine methodological advances with applied research tailored to domain-specific requirements, including limited computational resources and reliance on synthetic data for fine-tuning and evaluation.

C. SOLUTION OBJECTIVES

The proposed OCR system is designed to automate the verification of handwritten IDs and their corresponding signatures. The primary objectives of the system are as follows:

- Automatic and efficient verification: Accurately identify and validate handwritten IDs, ensuring correct pairing with their corresponding signatures.
- Performance evaluation: Generate detailed reports using predefined metrics to summarize verification outcomes, including the number of successfully verified IDs and those that failed verification.

- **Data privacy compliance:** Ensure compliance with the GDPR by employing synthetic datasets that simulate handwritten petition forms.

The system must process petition forms in which each page is treated as an image instance containing handwritten IDs and signatures. The output must include the successfully verified IDs and those flagged for manual review.

To ensure the system aligns with both procedural and technical requirements of the Parliament, the following constraints must be met:

- **Number of signatories:** According to parliamentary regulations, popular legislative initiatives must include at least 15,000 signatures, or signatures from 50% of the electorate in an island constituency if the initiative pertains solely to that region.
- **Document format:** Each page must follow a predefined tabular layout designed to record handwritten IDs and signatures. These documents are provided in PDF format and processed as images, simulating scanned, human-completed forms to ensure compatibility with OCR models.
- **ID format:** Each ID must conform to the official format specified by the Spanish Home Office [5], consisting of eight numeric digits followed by a single alphabetic character (see Figure 1).
- **Signature requirement:** Each signatory must provide a handwritten signature alongside the ID number.
- **Transformer-based architecture:** The system must rely exclusively on transformer-based models designed to be efficient and viable for use in resource-constrained environments typical of parliamentary institutions.
- **Dataset preparation:** Two distinct datasets must be dynamically generated to support system development and evaluation: (1) the ID Dataset, which simulates handwriting styles for IDs to facilitate TrOCR model fine-tuning and (2) the Form Document (FD) Dataset, representing the written petition and containing a minimum of 15,000 representative samples of handwritten IDs and signatures. This dataset is designed for system evaluation while ensuring compliance with GDPR. Both datasets are generated on demand based on user-defined parameters, such as the number of IDs for the ID dataset and the number of pages and signatories per page for the FD dataset. This approach enables flexible dataset creation tailored to different evaluation scenarios.

D. SCALABILITY AND ADAPTABILITY TO DIFFERENT JURISDICTIONS AND INSTITUTIONAL SETTINGS

The proposed multi-task transformer-based OCR pipeline is designed to be scalable and adaptable across different legal and parliamentary contexts. The system can handle variations in document layouts, handwritten identifiers, and signature formats by leveraging its end-to-end architecture, which simultaneously detects, recognizes, and pairs IDs and signatures.

To facilitate adaptation to new jurisdictions or institutions, synthetic dataset generation can be customized to reflect local document structures, handwriting styles, and identifier conventions. This allows the model to be fine-tuned effectively without requiring access to sensitive personal data, ensuring GDPR-compliant training and evaluation.

Additionally, the system's modular design supports integration into diverse institutional workflows, allowing automated verification pipelines to be deployed with minimal changes to existing processes. These features make the pipeline suitable for large-scale deployment in multiple legal and parliamentary environments, ensuring consistent performance while respecting local data protection and operational requirements.

IV. MATERIAL AND METHODS

The proposed OCR system is structured as a modular pipeline designed to support the automation and scalability required for processing large volumes of written petitions. As illustrated in Figure 3, the pipeline comprises three main modules:

- **Dataset preparation:** Described in Section IV-A, this module performs two key functions: (1) generating the ID dataset and (2) generating the FD dataset.
- **Data extraction:** This core module consists of three submodules: text detection, text recognition, and signature detection. The text detection submodule identifies the bounding boxes corresponding to handwritten IDs and signatures within the table image and the coordinates of table rows and columns. The text recognition submodule extracts ID values from the detected regions, while the signature detection submodule verifies whether a handwritten signature accompanies each detected ID.
- **Confirmation and Results:** Once an ID is recognized, it is validated against the official format specified by the Spanish Home Office. Finally, the system generates a report summarizing the results based on predefined performance metrics.

A. DATASET PREPARATION

A key requirement for developing and testing the proposed transformer-based OCR system is access to datasets containing annotated handwritten IDs and corresponding signatures. However, obtaining such datasets poses significant challenges, as they involve personal data and are subject to strict restrictions under the GDPR. Moreover, the dataset must include diverse and well-annotated samples to support effective ID recognition. Additionally, while the TrOCR model has been pre-trained on printed text and fine-tuned on English handwritten text, it has not been specifically fine-tuned for handwritten IDs.

To address these challenges and reduce the time and effort required to prepare such data, we propose a method for automatically generating synthetic annotated datasets of varying sizes. This approach ensures the diversity needed for fine-tuning the recognition model and evaluating its

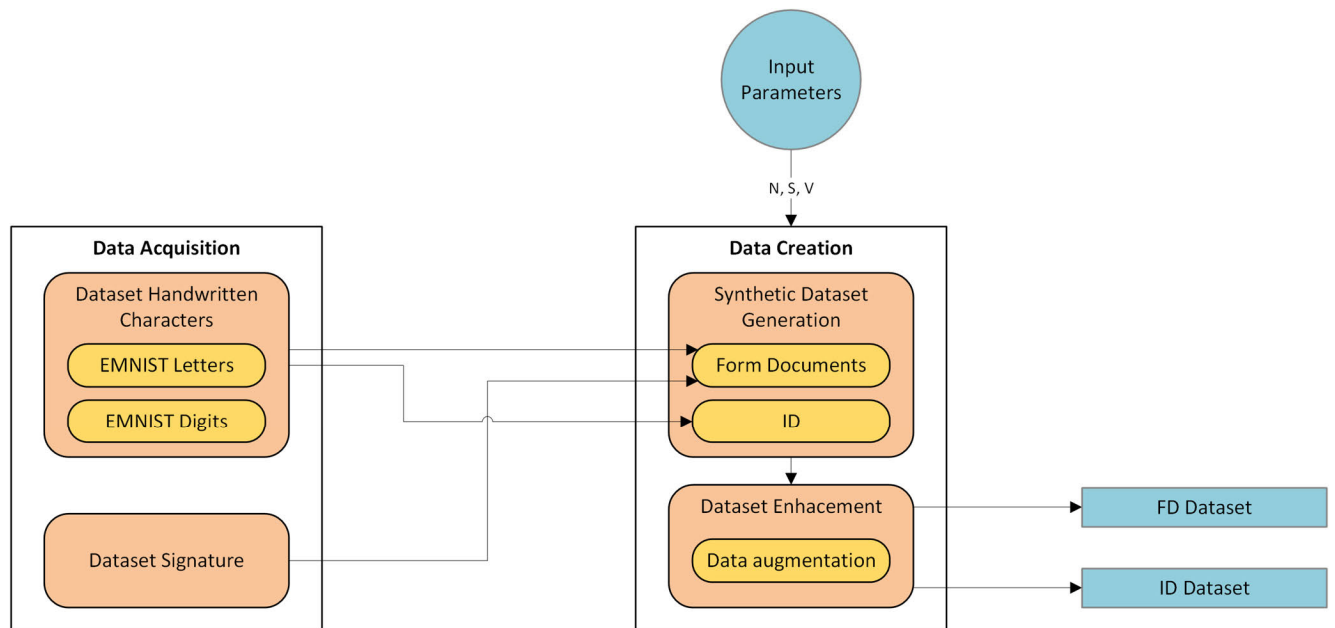


FIGURE 4. Synthetic Dataset Preparation Architecture. The figure illustrates the architecture for preparing dynamic synthetic datasets, where N represents the number of pages with handwritten IDs and signatures, S denotes the number of signatories per page for the FD dataset, and V indicates the number of handwritten IDs to be generated for the ID dataset.

performance, all while remaining GDPR-compliant. Our method for creating datasets involves extracting and merging annotated handwritten characters, including digits and letters, from the most relevant sources, as well as applying data augmentation techniques to further enhance their diversity and variety.

The dataset preparation module involves two key stages, as illustrated in Figure 4:

- **Data Acquisition:** This stage involves identifying and collecting handwritten characters and signatures.
- **Data Creation:** In this stage, the synthetic dataset generation process and the dataset enhancement process are applied to create two datasets: the ID Dataset and the FD Dataset. The ID Dataset is intended for fine-tuning the TrOCR model and requires, as an input parameter, the number of handwritten IDs to be generated (denoted as V). The FD Dataset simulates written petition documents and takes as input the number of pages (denoted as N) and the number of signatories per page (denoted as S), each containing both handwritten IDs and signatures. During the synthetic dataset generation process, ID images are automatically produced according to the specified parameters V and N , using unique combinations of eight digits followed by one letter, following the format established by the Spanish Home Office. In the FD Dataset, each generated ID image is paired with a corresponding synthetic signature image. Subsequently, the dataset enhancement process is applied. This involves data augmentation techniques aimed at increasing variability in handwritten styles across both datasets.

The components of each stage in the dataset preparation module are detailed below.

1) HANDWRITTEN CHARACTER ACQUISITION

The aim focuses on identifying the most appropriate datasets containing handwritten digits and letters to support the generation of synthetic ID images. The EMNIST dataset was selected due to its suitability as a benchmark for handwritten character recognition tasks [31]. Derived from the NIST Special Database 19 (SD19) [32], EMNIST includes over 800,000 manually verified and labeled samples contributed by nearly 3,700 writers [33], converted into a standardized 28×28 -pixel image format. This ensures a diverse and representative collection of handwritten characters, which is essential for constructing a robust dataset that captures natural variation in handwriting styles.

The EMNIST dataset provides six predefined subsets: ByClass, ByMerge, Letters, Digits, and MNIST. Some subsets, such as ByClass and ByMerge, exhibit class imbalance and are thus less suitable for model fine-tuning. Therefore, the Letters and Digits splits were selected due to their balanced class distributions. The Letters split comprises 26 alphabetic classes (a–z), combining upper and lowercase letters into a single set with a total of 145,600 samples. The Digits split includes 10 numeric classes (0–9), totaling 280,000 samples. These characteristics make EMNIST an ideal source for generating synthetic data in both the ID and FD datasets.

2) HANDWRITTEN SIGNATURE ACQUISITION

The objective is to identify and collect handwritten signature image samples to accompany each ID in the FD dataset. For

this purpose, a publicly available dataset by Suresh et al. [34], developed for handwritten signature recognition using deep learning, was selected. The dataset contains 2,500 signature images from 25 unique individuals, with 100 signatures per individual, each provided in a 224×224 -pixel image format.

3) SYNTHETIC DATASET GENERATION

The synthetic dataset generation process, the first step of the data creation phase, produces two distinct datasets: the ID Dataset and the FD Dataset. The FD Dataset is generated in PDF format, with each page containing a table image consisting of S rows and two columns, one for handwritten IDs and one for corresponding handwritten signatures, representing the specified number of signatories per page.

In accordance with the format defined by the Spanish Home Office, each ID consists of a unique combination of eight digits followed by one letter. To simulate this structure, ID images are constructed by concatenating randomly sampled digits from the EMNIST Digits dataset (280,000 samples; see Section IV-A1) and a letter from the EMNIST Letters dataset (145,600 samples). These components are combined to form a complete ID image that adheres to the official Spanish ID generation algorithm. For the ID Dataset, the process takes as input the number of IDs to be generated (denoted as V) and outputs a dataset in image format, with each entry representing a complete ID image.

4) SYNTHETIC DATASET ENHANCEMENT

The synthetic dataset enhancement process, the second step in the data creation phase, aims to increase the variability and realism of the generated ID images by applying data augmentation techniques. These techniques produce more diverse and representative handwritten samples for each ID generated in the previous step (see Section IV-A3). Four image transformations are applied to both the FD and ID datasets: random rotations (ranging from -5° to 5°), Gaussian blurring, image downscaling, and upscaling.

Table 2 presents the algorithm used to generate the FD dataset. This algorithm creates a specified number of pages (N), each containing a table image with S rows corresponding to signatories. Each signatory is assigned a synthetic ID composed of eight digits and a control letter, and is paired with a corresponding handwritten signature.

The algorithm for generating the ID dataset follows a similar approach but only requires the number of ID images to be generated as input. Figure 5 shows representative examples of ID and signature images extracted from the FD dataset.

B. DATA EXTRACTION

This module comprises three submodules, as illustrated in Figure 3: text detection, text recognition, and signature detection. Each submodule is implemented using a transformer-based model. The following subsections describe each submodule.

TABLE 2. Algorithm for synthetic FD dataset generation.

Input: number of pages N , number of signers S	
Output: PDF form document dataset FD	
1:	Initialize $FD \leftarrow \{\}$
2:	for $n = 1$ to N do
3:	Initialize $T[S] \leftarrow \{\}$ // Signatories Table
4:	for $s = 1$ to S do
5:	Initialize $ID \leftarrow \{\}$ // Identification Number Image
6:	Initialize $F \leftarrow \{\}$ // Signature Image
7:	$ID \leftarrow$ Image of eight handwritten numbers randomly generated from EMNIST Digits
8:	$ID \leftarrow$ Concat a control letter generated from EMNIST Letters, following the Spanish ID algorithm
9:	$ID \leftarrow$ Apply random image transformations: rotations (from -5 to 5 degrees), Gaussian blurring, image downscaling, and upscaling
10:	$F \leftarrow$ Randomly generated handwritten signature
11:	$T[S] = [ID, F]$
12:	end for
13:	$FD[n] = T$
14:	end for
15:	Return FD

1) TEXT DETECTION

The text detection submodule identifies the locations of handwritten IDs and signatures within the FD dataset. Its primary objective is to localize all ID and signature regions on each page image.

Since each page in the FD dataset follows a table structure, with rows corresponding to signatories and two columns representing ID and signature entries (see Section IV-A), this submodule takes each page image as input and outputs the coordinates of the bounding boxes that enclose the relevant regions. This step is essential, as the subsequent submodule requires the extracted region containing the sequence of tokens that represent the ID to be recognized.

To perform this task, a transformer-based object detection method is employed. Specifically, the Table Transformer model proposed by Smock et al. [18] is used. This model, trained on the PubTables 1M dataset, which comprises one million annotated tables, detects and interprets the structure of tables in document images. Based on this structure, the bounding boxes for each ID and signature are computed by iterating through the rows of each table.

For every bounding box identified, two image segments are extracted from the input image: one containing the ID and the other the corresponding signature. These segments are then forwarded to the subsequent submodules for ID recognition and signature verification.

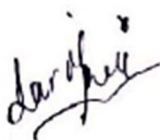
D.N.I	Firma
8302082PM	
24078370S	
44043318C	
82732722C	

FIGURE 5. Examples of four synthetically generated handwritten IDs and corresponding signatures for four signatories, extracted from the FD dataset. D.N.I. refers to “ID” and Firma means “signature” in Spanish.

2) TEXT RECOGNITION

The text recognition submodule is implemented as an end-to-end text recognition transformer model, based on the TrOCR architecture proposed by Li et al. [23]. TrOCR employs an image transformer as the encoder and a text transformer as the decoder. The authors present three model variants: small, base, and large, containing 62 million, 334 million, and 558 million parameters, respectively. This range allows a trade-off between computational efficiency and representational capacity, enhancing the model’s ability to capture fine-grained image details. While the large model achieves the highest performance, it is also the most computationally demanding.

The pretrained TrOCR models were initially trained on large-scale synthetic datasets, comprising hundreds of millions of printed text line images. These models were subsequently fine-tuned on the IAM Handwritten dataset, which is widely used for handwritten text recognition, and the SROIE dataset, which contains over one thousand scanned receipt images.

For this work, we adopt the large, fine-tuned version of the TrOCR model, initially trained on the IAM Handwritten dataset, to develop the text recognition submodule. This model integrates a BeiT transformer [35] as the encoder and RoBERTa [25] as the decoder, enabling image-to-text sequence processing. To adapt the model for recognizing

handwritten IDs, we further fine-tuned it using our ID datasets. As a result, in the fine-tuned model, the encoder processes patches from ID images to extract relevant features, while the decoder generates the corresponding text sequence for the ID.

3) SIGNATURE DETECTION

Signature detection within the predefined bounding boxes is approached as a zero-shot image classification task. To accomplish this, we adapted the SigLIP (Sigmoid Loss for Language-Image Pre-training) transformer-based model, originally designed for vision and language tasks, to the problem of signature detection. SigLIP processes individual image-text pairs directly without requiring a global normalization based on pairwise similarities [36]. Consequently, our task is framed as a binary classification problem in which images are classified as either “signature,” indicating the presence of a signature within the bounding box, or “no signature,” indicating its absence.

C. CONFIRMATION AND RESULTS

This module is responsible for verifying the validity of each ID recognized by the data extraction module. It ensures the accuracy and compliance of the recognized data through a structured process. The module consists of three submodules, which are described in detail below:

1) POST-PROCESSING

The decoder of the TrOCR model may produce incorrect characters at certain positions within the text sequence that do not conform to the structural definition of an ID (a unique combination of eight digits followed by one letter). Typical errors include the introduction of non-alphanumeric characters such as punctuation marks (e.g., commas, colons, dashes, periods), unexpected whitespace, substitution of uppercase letters for lowercase letters, and the misclassification of numbers as letters or vice versa. To address these issues, the post-processing submodule refines the ID text sequences through the following steps:

- Removal of non-alphanumeric characters (step 1): All non-alphanumeric characters, including whitespace and punctuation marks, are filtered out.
- Capitalization correction (step 2): All lowercase letters are converted to their uppercase equivalents.
- Substitution of visually similar letters with numbers (step 3): When a digit is expected in the first eight positions but a letter is detected, it is replaced by the most visually similar number based on the following mapping: (“A” → “4”, “B” → “3”, “G” → “6”, “I” → “1”, “O” → “0”, “P” → “9”, “S” → “5”, “T” → “7”, and “Z” → “2”). Additionally, certain non-alphanumeric characters are replaced with similar digits: (“/”, “(”, “)”, “!” → “1”, and “&” → “8”).
- Substitution of visually similar numbers with letters (step 4): When a letter is expected in the last position

of the text sequence but a digit is detected, it is replaced by the most visually similar letter according to the following mapping: (“0” → “O”, “7” → “T”, “2” → “Z”, “3” → “B”, “4” → “A”, “5” → “S”, and “6” → “G”, and “9” → “P”).

2) INTEGRITY

Following the corrections performed by the postprocessing submodule, the integrity submodule is responsible for ensuring the accuracy and integrity of each ID. This process encompasses several key functionalities aimed at verifying the conformity of the IDs to their structural and algorithmic requirements:

- **Alignment with the Structural Definition of an ID:** This functionality verifies that each ID adheres to the structural format, consisting of a unique combination of eight numerical digits followed by one alphabetic character.
- **Control Character Validation:** This step ensures that the ID’s control character is defined according to the algorithm established for Spanish IDs, maintaining the integrity of the identification process.
- **Anomaly Detection:** This process identifies potential issues, such as duplicate IDs, to ensure that all IDs are unique.

3) REPORT

The report submodule is responsible for generating summaries of the validation outcomes. Its primary functions include:

- **Error Reporting:** This component lists all IDs that failed validation.
- **Performance Metrics:** This component evaluates the accuracy and reliability of the text recognition submodule used for ID recognition. By employing both ML and NPL metrics, it assesses the effectiveness of the recognition process. These metrics contribute to a comprehensive evaluation of the system’s overall performance.

V. RESULTS

The transformer-based OCR system was evaluated as a prototype for potential real-world implementation within the Parliament of the Canary Islands. The VVT framework [29] assessed the system’s performance and practical utility. The experimental setup and the results of the verification and validation process used in the evaluation are described below.

A. EXPERIMENTAL SETUP

This subsection presents the experimental setup, including the computing environment, dataset preparation, evaluation metrics, baseline model, and the step-by-step workflow followed to implement and assess the transformer-based OCR system.

1) ENVIRONMENT

The system was implemented using Python 3.13, with Pytorch as the deep learning framework, and the large version of the pre-trained TrOCR model fine-tuned on the IAM handwritten dataset [23]. It was further fine-tuned on our custom ID datasets to enhance the model’s ability to recognize handwritten IDs, as detailed in Section IV-A.

All training and evaluation tasks were conducted on a system equipped with an Intel Core™ i7-10700 CPU (2.9 GHz) and 32 GB of RAM, aiming to assess the system’s performance and feasibility in resource-constrained parliamentary environments. For comparison purposes, the baseline model was evaluated on a more powerful setup comprising an Intel Core™ i7-11800H CPU (2.3 GHz), an NVIDIA GeForce RTX 3050 GPU, and 32 GB of RAM.

2) DATASETS

Two synthetic datasets were generated, as described in Section IV-A:

- **ID Dataset:** Five fine-tuning sets of varying sizes ($V = 100, 1,000, 2,000, 3,000$, and $4,000$ samples) were created, each containing annotated ID samples to evaluate the TrOCR model’s robustness as the training set size increases. Training was conducted for three epochs using a learning rate of 5.0×10^{-6} .
- **FD Dataset:** This dataset simulates the Parliament’s real-world operational procedures (see Sections III-C) and serves as a test set to evaluate the performance of three transformer-based models: the Table Transformer for text detection, TrOCR for ID recognition, and SigLIP for signature detection. In accordance with procedural requirements, popular legislative initiatives must include a minimum of 15,000 signatures. To reflect this, the dataset was generated using input parameters of $N = 1,500$ pages and $S = 10$ signatories per page, yielding 15,000 synthetic signatory records. This version is referred to as FDD-S. Additionally, a smaller subset was created to evaluate TrOCR’s performance on authentic handwriting samples. This subset consists of $N = 50$ pages with $S = 10$ signatories per page, resulting in 500 real-world signatory records. Referred to as FDD-R, it is used to assess the model’s generalization capabilities to real-world data. Importantly, both the 15,000 synthetic samples and the 500 real-world samples were reserved exclusively for testing and were not used during any stage of the fine-tuning process.

3) METRICS

The performance was evaluated from both NLP and ML perspectives:

NLP perspective: Focused on character-level accuracy, this evaluates fine-grained errors in text recognition by comparing predicted characters to ground truth labels. The key metrics included:

Precision: measures the ratio of correctly predicted positive characters to the total predicted positives. It quantifies how many of the ID's characters predicted by the model are correct:

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

where TP (true positives) corresponds to the number of correctly recognized characters and FP (false positives) to the number of incorrectly recognized characters.

Accuracy: represents the ratio of correctly predicted characters to the total number of characters in the actual ID:

$$Accuracy = \frac{TP}{Total\ number\ of\ characters\ in\ the\ actual\ ID} \quad (2)$$

Recall: measures the ratio of correctly predicted characters to the total number of actual positive characters. It focuses on how many of the real positive instances were correctly identified by the model:

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

where FN (false negatives) represents the number of unrecognized characters.

F1-Score: The harmonic mean of precision and recall:

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

Character Error Rate (CER): measures the number of character-level errors, including insertions, deletions, and substitutions required to transform the model's prediction into the correct ID:

$$CER = \frac{S + D + I}{N} \quad (5)$$

where S corresponds to the number of substitutions, D to the number of deletions, I to the number of insertions, and N to the total number of characters in the actual ID.

ML Perspective: Focused on ID-level accuracy, this evaluates the system's ability to correctly recognize complete IDs as whole entities. The evaluation metrics included:

Precision: measures the proportion of correctly recognized IDs among all IDs predicted as correct:

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

where TP (true positives) corresponds to the number of IDs correctly identified as positive and FP (false positives) to the number of IDs incorrectly identified as positive.

Accuracy: represents the proportion of all correct predictions (positive and negative) relative to the total predictions:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

where TN (true negatives) refers to the number of IDs correctly identified as negative, while FN (false negatives) represents the number of valid IDs that the model incorrectly predicts as not matching the ground truth.

Recall: measures the proportion of correctly recognized IDs out of all IDs that should have been recognized:

$$Recall = \frac{TP}{TP + FN} \quad (8)$$

F1-Score: is the harmonic mean of precision and recall:

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (9)$$

4) BASELINE MODEL

We use the CRNN model [37] as the baseline model, as it has traditionally been the state of the art for text recognition. The CRNN model consists of convolutional layers for image feature extraction, recurrent layers for sequence modeling and final frame label prediction, and a transcription layer to convert frame predictions into the final label sequence.

5) IMPLEMENTATION WORKFLOW

To ensure reproducibility and provide methodological clarity, this subsection presents the step-by-step experimental workflow used to conduct the evaluation. The workflow is illustrated in a flow diagram (see Figure 6), where each box represents a numbered step in the sequence, providing a visual overview of the entire process from dataset generation to model evaluation. Post-processing configurations are defined as follows: NPS refers to no post-processing, PS1 refers to applying only steps 1 and 2, and PS2 refers to applying the full set of post-processing steps described in Section IV-C.1. The workflow is summarized in the following sequential steps:

Step 1) Initialize input parameter V, N, S

Step 2) Dataset preparation

Step 2.1) For each value of V , generate an ID dataset for training purposes.

Step 2.2) Generate the datasets FDD-S and FFD-R for testing purposes.

Step 3) Model fine-tuning

Step 3.1) Fine-tune the TrOCR model using the ID datasets from Step 2.1 to obtain different models for performance evaluation.

Step 4) Text detection and extraction evaluation

Step 4.1) Apply the transformer-based extraction model to identify IDs and signatures in each tabular line of FFD-S.

Step 4.2) Apply the transformer-based extraction model to locate IDs and signatures in each tabular line of FFD-R.

Step 4.3) Compute metric results for FFD-S and FFD-R to verify text detection accuracy.

Step 5) Text recognition evaluation using FFD-S and FFD-R

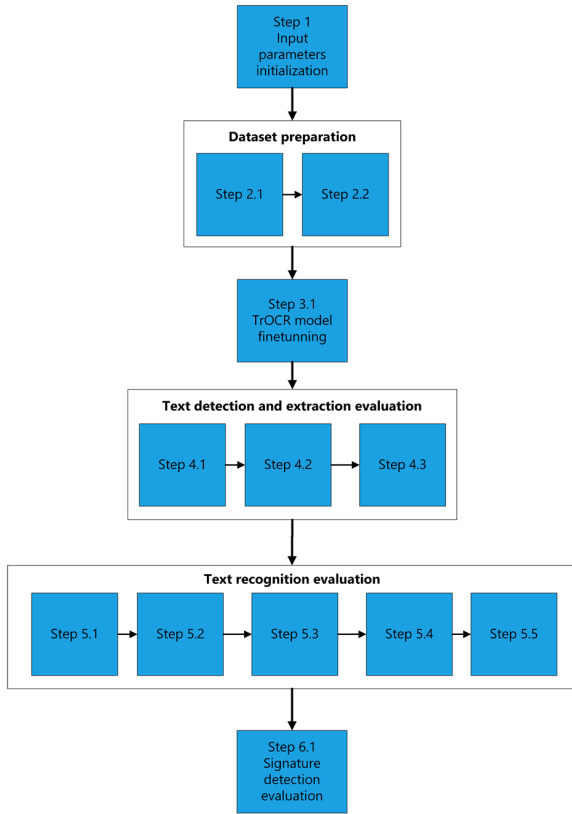


FIGURE 6. Flow diagram of the experimental workflow described in Section V-A5. Each box represents a numbered step from the step-by-step procedure, covering key stages such as input parameter initialization, dataset preparation, model fine-tuning, and metrics computation. Arrows indicate the sequence in which steps are executed. This diagram provides a visual overview of the workflow, clarifying the implementation process.

Step 5.1) Compute NLP and ML metrics using the original TrOCR model for the postprocessing configurations NPS, PS1, and PS2.

Step 5.2) Compute NLP and ML metrics using each fine-tuned TrOCR model from Step 3.1 for the same postprocessing configurations.

Step 5.3) Select the best-performing finetuned TrOCR model based on the results from Step 5.2.

Step 5.4) Compute NLP and ML metrics for the baseline model.

Step 5.5) Compare the results from Steps 5.3 and 5.4 to identify the best overall model.

Step 6) Signature detection evaluation

Step 6.1) Compute metrics for signature detection using the SigLIP transformer-based model on FFD-S and FFD-R.

Figure 6 illustrates the flow diagram of the step-by-step workflow, providing a visual representation of the sequence of steps and operations for each stage.

B. VERIFICATION

The verification process evaluates whether the prototype meets the requirements, objectives, and dataset preparation criteria defined in Section III.

- **Requirements Verification:** All specified requirements were fully implemented and validated, including the number of signatories, the form document template, the ID format, the inclusion of handwritten signatures, and the transformer-based processing approach. This validation was carried out by researchers of ULL and the CIO of the Parliament of the Canary Islands, ensuring compliance with the predefined specifications.
- **Objectives Verification:** All project objectives were successfully achieved. The PoC delivered an automated solution for HSV applied to written petitions. It also generated synthetic datasets of various sizes to fine-tune the TrOCR model and evaluate the system's performance, while adhering to data protection requirements under the GDPR. User involvement was minimal, limited to entering parameters N (number of pages), S (number of signatories per page), and V (dataset size).
- **Dataset Preparation:** The dataset preparation process followed the method outlined in Section IV-A. Synthetic data generation techniques produced realistic representations of handwritten signatures, ensuring strict compliance with data privacy regulations.

C. VALIDATION

The validation process assessed the performance of the data extraction module, focusing on three submodules: text detection, text recognition, and signature detection.

1) TEXT DETECTION RESULTS

The Table Transformer model [18] was evaluated to ensure the accurate calculation of bounding box coordinates for each ID and signature across all rows of the table structure on every page of the FD dataset. The model successfully detected the corresponding bounding boxes for all pages processed.

2) TEXT RECOGNITION RESULTS

To evaluate the performance of the TrOCR model [23] in recognizing IDs, we conducted six distinct experiments. These evaluations incorporated both NLP and ML metrics. The experiments were designed to analyze the impact of varying fine-tuning dataset sizes and post-processing configurations. All experiments were carried out using both the FFD-S (synthetic dataset) and FDD-R (real-world dataset).

To investigate how the size of the fine-tuning dataset influences model performance, we generated five datasets of sizes $V = 100, 1,000, 2,000, 3,000$, and $4,000$. Two post-processing configurations were evaluated: (1) applying only steps 1 and 2 (PS1), and (2) applying the full set of post-processing steps (PS2), as described in Section IV-C1. Based on these factors, the following experiments were carried out:

- 1) **Original Model (No Fine-tuning, No Post-processing):** The first experiment evaluated the original model without any fine-tuning or post-processing.
- 2) **Original Model (No Fine-tuning, Post-processing Applied):** This experiment assessed the original model

TABLE 3. NLP and ML metric results for the TrOCR model without fine-tuning on ID datasets. FFD-S: FD dataset with N = 1,500 pages containing synthetic data. FFD-R: FD Dataset with N = 50 pages containing real human handwriting data. NPS: No post-processing steps applied. PS1: Post-processing with step 1 and step 2. PS2: Comprehensive post-processing, including step 1, step 2, step 3.

Model	Postprocessing Method	NPL metrics					ML metrics			
		Accuracy	Precision	Recall	F1	CER	Accuracy	Precision	Recall	F1
TrOCR FDD-S	NPS	69.11	70.02	86.65	77.45	30.89	8,08	96,21	8,09	9,71
TrOCR FDD-S	PS1	83.36	85.54	84.83	85.18	16.64	51.06	86.01	51.88	64.72
TrOCR FDD-S	PS2	84.38	86.58	84.67	85.61	15.62	69.96	92.26	73.26	81.51
TrOCR FDD-R	NPS	56.08	57.01	73.62	64.25	43.92	5.08	100	5.08	9.66
TrOCR FDD-R	PS1	70.31	72.52	71.82	72.16	29.69	38.04	72.98	38.86	50.71
TrOCR FDD-R	PS2	71.36	73.54	71.67	72.59	28.64	46.93	69.24	50.23	58.22

without fine-tuning but with post-processing. Two variations were tested: (1) applying only steps 1 and 2, and (2) applying all four post-processing steps.

- 3) Fine-tuned Model (No Post-processing): In this experiment, the model was fine-tuned on custom ID datasets of varying sizes ($V = 100$ to $4,000$) without applying any post-processing steps.
- 4) Fine-tuned Model (Post-processing Applied with Steps 1 and 2): This configuration combined fine-tuning with post-processing steps 1 and 2 to assess their joint effect.
- 5) Fine-tuned Model (Post-processing Applied with Steps 1, 2, 3, and 4): The model was fine-tuned on datasets of varying sizes and subjected to all four post-processing steps.
- 6) Comparison with Baseline Model: The best-performing configuration from the above experiments was compared against the baseline model to evaluate performance improvements.

Table 3 presents the results of experiments 1 and 2. The key findings are as follows: the results reveal a performance gap between synthetic and real handwriting data, with lower accuracy and F1 scores observed on FFD-R across all evaluation metrics. For both datasets, the application of post-processing techniques proves essential. Without post-processing (NPS), performance remains limited, particularly under the ML-based evaluation, as this requires the correct recognition of complete ID strings as unified entities. Applying basic post-processing (PS1) substantially improves performance, especially for the synthetic dataset, where the F1 score increases from 9.71% to 64.72%. The most comprehensive post-processing configuration (PS2) yields the highest results across both datasets, achieving an F1 score of 81.51% on FFD-S and 58.22% on FFD-R. These findings underscore the effectiveness of post-processing in mitigating character-level errors, compensating for the absence of fine-tuning.

Table 4 summarizes the results of experiment 3, which involved fine-tuning the model on progressively larger

datasets ($V = 100$ to $4,000$). Both NLP and ML evaluation metrics show incremental improvements as the dataset size increases, particularly up to $V = 3,000$. Results reported in Tables 5 and 6 further indicate that post-processing has no effect when applied after fine-tuning, suggesting that fine-tuning itself is the principal factor contributing to performance improvements. For instance, the NLP-based accuracy on the FDD-S dataset improved from 97.74% at $V = 100$ to 98.91% at $V = 3,000$. However, the outcomes remained consistent regardless of whether post-processing was applied. Similar patterns were observed in the ML-based metrics across all configurations, reinforcing the conclusion that fine-tuning is the dominant contributor to the observed gains.

Table 7 presents a comparative evaluation of the CRNN and fine-tuned TrOCR models on both FDD-S and real FDD-R datasets using NLP and ML evaluation metrics. The results demonstrate a performance gap between the two models across all metrics and datasets. On the synthetic dataset (FDD-S), the CRNN model achieved a performance, with an NLP F1 score of 81.21% and a Character Error Rate (CER) of 29.76%. However, its performance on ML metrics was notably poor, achieving an F1 score of only 12.6%. In contrast, the TrOCR model fine-tuned on 3,000 samples reached an NLP F1 score of 99.39% with a CER of just 1.09% and demonstrated a superior ML F1 score of 95.30%. The same trend holds for the real handwriting dataset (FDD-R). For this dataset, CRNN obtained an NLP F1 score of 66.22% and an ML F1 score of 7.39%, indicating limited generalization to authentic handwritten input. Meanwhile, TrOCR fine-tuned on 3,000 samples outperformed CRNN by a wide margin, achieving an NLP F1 score of 84.71% and an ML F1 score of 88.96%.

These results indicate that the transformer-based TrOCR model, when fine-tuned with a sufficiently large synthetic dataset, outperforms the traditional CRNN architecture in both synthetic and real-world scenarios.

TABLE 4. NLP and ML metric results for the TrOCR model, fine-tuned on ID datasets of varying sizes ($V = 100, 1,000, 2,000, 3,000, 4,000$) with no post-processing steps applied. FFD-S: FD dataset with $N = 1,500$ pages containing synthetic data. FFD-R: FD Dataset with $N = 50$ pages containing real human handwriting data.

Model	ID (V)	NPL metrics					ML metrics			
		Accuracy	Precision	Recall	F1	CER	Accuracy	Precision	Recall	F1
TrOCR FDD-S	100	97.74	98.84	98.84	98.84	2.26	80.00	88.89	88.89	88.89
TrOCR FDD-S	1000	98.02	98.86	98.84	98.84	1.98	81.96	88.44	88.86	88.64
TrOCR FDD-S	2000	98.11	98.86	98.86	98.86	1.89	82.32	88.44	88.84	88.63
TrOCR FDD-S	3000	98.91	99.89	98.91	99.39	1.09	90.48	96.04	94.59	95.30
TrOCR FDD-S	4000	98.14	98.82	98.84	98.82	1.86	89.32	91.02	91.24	91.12
TrOCR FDD-R	100	83.64	84.36	84.52	84.43	16.36	73.22	80.02	80.00	80.00
TrOCR FDD-R	1000	84.00	84.34	84.36	84.34	16.00	74.21	84.06	82.24	83.14
TrOCR FDD-R	2000	84.06	84.51	84.48	84.49	15.94	78.44	86.02	86.44	86.22
TrOCR FDD-R	3000	84.31	84.58	84.86	84.71	15.69	80.04	88.96	88.96	88.96
TrOCR FDD-R	4000	83.98	84.52	84.32	84.41	16.02	80.01	86.02	86.24	86.12

3) SIGNATURE DETECTION RESULTS

The SigLIP transformer-based model [36] was evaluated as a binary classification task to confirm the presence of signatures in conjunction with the IDs detected. For all pages processed from the FD dataset, the model achieved perfect accuracy and precision, with 100% in both metrics. Consequently, the model successfully predicted both the presence and absence of signatures across all instances.

VI. DISCUSSION

The key findings obtained from experiments one to six for text recognition are as follows:

- **Post-processing:** Post-processing steps had a direct impact on model performance, particularly when the TrOCR model was not fine-tuned. The experiments carried out showed how each step contributed to the observed performance and clarified the relationship between post-processing, fine-tuning, and evaluation metrics. Without fine-tuning, the TrOCR decoder generated incorrect characters in positions that did not conform to the structural definition of an ID (a unique combination of eight digits followed by one letter). Typical errors included insertion of non-alphanumeric characters (commas, colons, dashes, periods), unexpected whitespace, substitution of uppercase with lowercase letters, and misclassification of numbers as letters or vice versa. These errors led to a decrease in both character-level (NLP) and ID-level (ML) metrics, as the model could predict individual characters but fail to reconstruct the entire ID string accurately.

Applying post-processing mitigated these errors. When all post-processing steps were applied, character-level errors were reduced and ID-level accuracy improved, yielding strong results even under frequent decoder errors. For instance, in non-fine-tuned conditions, ML F1 improved from 9.71% to 81.51% on synthetic data. This demonstrates that post-processing can compensate for decoder errors and enforce structural consistency, which is critical for practical applications where complete ID recognition is necessary for legal or operational purposes.

However, once the model was fine-tuned, the benefits of post-processing became negligible. Fine-tuning on custom datasets inherently adapted the model to the structural constraints of IDs, reducing the frequency of decoder errors and achieving high performance across both NLP and ML metrics. This indicates that while post-processing is a useful corrective measure for non-fine-tuned models, fine-tuning is the dominant factor in achieving optimal performance.

As a result, the experiments show that post-processing is essential for improving results in early-stage or non-fine-tuned models, particularly in structured sequence recognition tasks, while fine-tuning ensures that the model generalizes well and minimizes reliance on post-processing corrections.

- **Fine-tuning:** Fine-tuning the TrOCR model on custom datasets, particularly those of larger size, significantly improved performance across both NLP and ML metrics, demonstrating the substantial impact of dataset scale on the model's ability to reconstruct complete

TABLE 5. NLP and ML metric results for the TrOCR model, fine-tuned on ID datasets of varying sizes ($V = 100, 1,000, 2,000, 3,000, 4,000$) with post-processing applied: step 1 and step 2. FFD-S: FD dataset with $N = 1,500$ pages containing synthetic data. FFD-R: FD Dataset with $N = 50$ pages containing real human handwriting data.

Model	ID (V)	NPL metrics					ML metrics			
		Accuracy	Precision	Recall	F1	CER	Accuracy	Precision	Recall	F1
TrOCR FDD-S	100	97.74	98.84	98.84	98.84	2.26	80.00	88.89	88.89	88.89
TrOCR FDD-S	1000	98.02	98.86	98.84	98.84	1.98	81.96	88.44	88.86	88.64
TrOCR FDD-S	2000	98.11	98.86	98.86	98.86	1.89	82.32	88.44	88.84	88.63
TrOCR FDD-S	3000	98.91	99.89	98.91	99.39	1.09	90.48	96.04	94.59	95.30
TrOCR FDD-S	4000	98.14	98.82	98.84	98.82	1.86	89.32	91.02	91.24	91.12
TrOCR FDD- R	100	83.64	84.36	84.52	84.43	16.36	73.22	80.02	80.00	80.00
TrOCR FDD- R	1000	84.00	84.34	84.36	84.34	16.00	74.21	84.06	82.24	83.14
TrOCR FDD- R	2000	84.06	84.51	84.48	84.49	15.94	78.44	86.02	86.44	86.22
TrOCR FDD- R	3000	84.31	84.58	84.86	84.71	15.69	80.04	88.96	88.96	88.96
TrOCR FDD- R	4000	83.98	84.52	84.32	84.41	16.02	80.01	86.02	86.24	86.12

IDs accurately. The improvements are particularly pronounced for synthetic data because fine-tuning allows the model to adapt to the structural characteristics of IDs and the statistical patterns of the generated handwriting. In contrast, real-world data (FDD-R) exhibited smaller performance gains with fine-tuning, reflecting the additional variability, irregularities, and noise inherent in authentic handwriting that are not fully captured by synthetic datasets. Nonetheless, even with this variability, fine-tuning enables the model to generalize to real-world scenarios, demonstrating its robustness for practical applications in parliamentary handwritten signature verification.

The results justify the role of fine-tuning in improving both character-level and ID-level recognition, highlight the importance of scaling training datasets, and show that transformer-based architectures can leverage fine-tuning to achieve high performance even under limited availability of real handwriting data.

- **Real-world Data Challenges:** While the results for both metrics from real-world data (FDD-R) were promising for the Parliament of the Canary Islands, with an accuracy of 84.31% for the NLP metric and 80.04% for the ML metric, the TrOCR model performed best on synthetic data (FDD-S), achieving 98.91% for NLP and 90.48% for ML metrics. This difference reflects a known limitation of synthetic datasets: although they

provide scalable, controllable, and GDPR-compliant training resources, they cannot fully capture the variability, irregularities, and document artifacts present in genuine handwriting [51].

Previous studies have shown that models trained exclusively on synthetic or constrained datasets may exhibit lower accuracy when applied to authentic handwriting, highlighting the inherent challenges of capturing the full diversity of writing styles [52–53]. In this context, the results of our experiments remain highly positive: despite being trained solely on synthetic data, the transformer-based system achieved competitive performance on the real-world dataset (FDD-R), with NLP and ML metrics of 84.31% and 80.04%, respectively. This demonstrates that the model generalizes to authentic inputs and confirms the robustness of the proposed pipeline. By validating both on FDD-S and FDD-R, the proof of concept establishes that transformer-based architectures can reliably support handwritten signature verification in parliamentary scenarios, combining accuracy with compliance to data-protection requirements.

- **CRNN vs TrOCR :** The CRNN baseline achieved moderate performance in character-level (NLP) metrics, with F1 scores of 81.21% on synthetic data (FDD-S) and 66.22% on real handwriting (FDD-R). This suggests that CRNN can recognize many individual characters, capturing local patterns in the handwritten IDs.

TABLE 6. NLP and ML metric results for the TrOCR model, fine-tuned on ID datasets of varying sizes ($V = 100, 1,000, 2,000, 3,000, 4,000$) with comprehensive post-processing applied: step 1, step 2, step 3, and step 4. FFD-S: FD dataset with $N = 1,500$ pages containing synthetic data. FFD-R: FD Dataset with $N = 50$ pages containing real human handwriting data.

Model	ID (V)	NPL metrics					ML metrics			
		Accuracy	Precision	Recall	F1	CER	Accuracy	Precision	Recall	F1
TrOCR FDD-S	100	97.74	98.84	98.84	98.84	2.26	80.00	88.89	88.89	88.89
TrOCR FDD-S	1000	98.02	98.86	98.84	98.84	1.98	81.96	88.44	88.86	88.64
TrOCR FDD-S	2000	98.11	98.86	98.86	98.86	1.89	82.32	88.44	88.84	88.63
TrOCR FDD-S	3000	98.91	99.89	98.91	99.39	1.09	90.48	96.04	94.59	95.30
TrOCR FDD-S	4000	98.14	98.82	98.84	98.82	1.86	89.32	91.02	91.24	91.12
TrOCR FDD-R	100	83.64	84.36	84.52	84.43	16.36	73.22	80.02	80.00	80.00
TrOCR FDD-R	1000	84.00	84.34	84.36	84.34	16.00	74.21	84.06	82.24	83.14
TrOCR FDD-R	2000	84.06	84.51	84.48	84.49	15.94	78.44	86.02	86.44	86.22
TrOCR FDD-R	3000	84.31	84.58	84.86	84.71	15.69	80.04	88.96	88.96	88.96
TrOCR FDD-R	4000	83.98	84.52	84.32	84.41	16.02	80.01	86.02	86.24	86.12

However, it performs poorly on ID-level (ML) metrics, with F1 scores of 12.6% on FDD-S and 7.39% on FDD-R, reflecting its difficulty in reconstructing entire IDs as unified entities. This limitation stems from the CRNN architecture, which combines convolutional layers for feature extraction with recurrent layers for sequence modeling but is less capable than transformer-based models at capturing long-range dependencies and global context across the full input sequence.

In contrast, the transformer-based TrOCR model leverages self-attention mechanisms to model relationships between all characters in an input sequence, resulting in higher performance across both NLP and ML metrics (FDD-S: NLP F1 99.39%, ML F1 95.30%; FDD-R: NLP F1 84.71%, ML F1 88.96%). These results justify the choice of TrOCR as the main model while showing that CRNN, despite its limitations in complete ID reconstruction, remains effective at character-level recognition, consistent with its role as a baseline in the original TrOCR study [23]. The comparison underscores the advantages of transformer architectures for handwritten signature verification, particularly when full ID recognition is critical for legal and operational purposes.

As a result, the experiments indicate that the TrOCR model achieves its best performance when fine-tuned on a sufficiently large synthetic dataset, with post-processing providing no additional improvements. By contrast, the CRNN model performs worse than TrOCR, particularly when handling real-world handwriting.

To ensure a comprehensive evaluation, the system was tested by the CIO of the Parliament of the Canary Islands.

These evaluations focused on performance metrics (outlined in Section V-C) and compliance with the requirements and objectives specified in Section III. The CIO's assessment revealed the following insights:

- Effectiveness of the transformer-based approach: The transformer-based architecture demonstrated effectiveness in automating handwritten signature verification. As detailed in Section V-C, the system achieved the target accuracy levels, fulfilling the objectives related to performance. Error rates, including false positives, false negatives, and misclassifications, were evaluated from both NLP and ML perspectives and were found to be comparable to human-level performance. While the system performed slightly better on synthetic data than on real handwriting samples, the results remain satisfactory, especially considering that real-world data were not available for training due to GDPR constraints. The model's ability to generalize to authentic inputs supports the viability of this approach in practical settings.
- Efficiency and minimal manual intervention: The HSV task has traditionally been performed manually, making it labor-intensive, time-consuming, and prone to human error. The PoC demonstrated its potential to improve efficiency by automating the process, thereby reducing the need for manual intervention.
- Scalability and compliance: The system dynamically generated synthetic datasets, ensuring scalability while adhering to GDPR requirements. This capability allows the solution to accommodate varying scenarios without compromising data protection standards.

TABLE 7. NLP and ML metric results for the CRNN model. FFD-S: FD dataset with N = 1, 500 pages containing synthetic data. FFD-R: FD dataset with N = 50 pages containing real human handwriting data.

Model	FFD	NPL metrics					ML metrics			
		Accuracy	Precision	Recall	F1	CER	Accuracy	Precision	Recall	F1
CRNN	FFD-S	70.24	70.88	88.77	81.21	29.76	12.80	10.82	15.11	12.6
TrOCR	3000	98.91	99.89	98.91	99.39	1.09	90.48	96.04	94.59	95.30
CRNN	FFD-R	55.21	55.82	72.73	66.22	44.79	7.42	5.39	11.78	7.39
TrOCR	3000	84.31	84.58	84.86	84.71	15.69	80.04	88.96	88.96	88.96

- Feasibility of the solution: The testing phase demonstrated that the proposed system is feasible for practical deployment, even when developed using CPU-based hardware. The solution leverages transformer-based models and performs for its intended tasks, such as text detection, text recognition, and signature detection. Despite the absence of GPU hardware, the system proved practical for deployment in parliamentary institutions with limited computational resources. Additionally, the use of synthetic datasets eliminates reliance on sensitive real-world data, addressing
- privacy concerns while supporting the fine-tuning of transformer models.
- Hardware limitations: While the solution was successfully implemented using general-purpose CPU-based equipment, the evaluation underscored the need for advanced hardware with GPU or TPU capabilities to fully leverage the potential of transformer-based models. Although the system operates efficiently for its current objectives, fine-tuning processes require significant computational power, which could be improved with GPU acceleration to enhance processing speed.
- Knowledge gaps in Public Institutions: The CIO noted that public institutions lack expertise in transformer-based models. This highlights the need for collaboration between universities, research centers, and public institutions to drive the adoption of AI-driven innovations.
- Adaptation to new programming paradigms: Public administration has traditionally relied on computing paradigms focused on algorithm design and programming, where the primary objective is to create new algorithms to solve problems directly. The shift from traditional computing to fine-tuning transformer models introduces a data-centric approach. Traditional software development focuses on deterministic, rule-based algorithms and uses general-purpose tools, whereas fine-tuning transformer models require specialized frameworks like PyTorch, along with significant computational resources (e.g., GPUs, TPUs). This new paradigm emphasizes iterative experimentation. Unlike traditional development, fine-tuning involves adapting pre-trained models to the specific task and addressing ethical concerns like algorithmic bias.

In summary, the testing process confirmed the system's ability to meet its intended objectives and validated its readiness for real-world implementation. Addressing hardware limitations and bridging institutional knowledge gaps will be essential for scaling the solution and fostering broader adoption of transformer-based models in legislative and other public sector applications.

VII. CONCLUSION

Transformer-based models represent the state of the art in NLP and CV. However, their adaptability to new, previously unseen data, real-world data scarcity, and high computational demands pose significant challenges when developing viable solutions within institutional settings. This paper presented a proof of concept OCR system to address the HSV problem in a parliamentary context. Developed and tested in collaboration with the Parliament of the Canary Islands, the proposed system meets procedural and technical requirements, demonstrating the feasibility of automating this task through a transformer-centric approach. It addresses challenges such as handwriting variability and compliance with data privacy regulations while evaluating the practicality of deployment in resource-constrained parliamentary environments. The key findings and contributions are as follows:

- Transformer-based adaptation: This work shows how transformer-based models, specifically the Table Extraction Model, TrOCR, and SigLIP, can be effectively integrated and adapted for HSV. The proposed system outperforms conventional CRNN frameworks in both synthetic and real-world scenarios.
- Privacy-compliant data generation: To overcome limitations on accessing available real handwritten data, synthetic datasets were generated to simulate realistic handwriting styles. These datasets comply with data protection regulations and enabled effective model training.
- Performance evaluation: System performance was assessed using a VVT framework and ML/NLP-based metrics. Results demonstrated strong performance when training and testing on synthetic data, achieving 98.91% character-level accuracy and 90.48% ID-level accuracy. When tested on real data, trained only on synthetic samples, the system achieved 84.31% and 80.04%

in character-level and ID-level accuracy, respectively. These findings validate the effectiveness of fine-tuning TrOCR on privacy-compliant datasets for ID recognition under real-world variability.

- Operational feasibility: The results confirm that it is possible to perform the HSV task with a transformer-centric approach with low computational costs, facilitating its adoption in real-life institutional settings. CIO feedback emphasized limited AI expertise in public institutions, underlining the need for stronger collaboration between academia and government.
- Institutional impact: According to the CIO, automating the HSV task for written petitions offers substantial advantages. It streamlines workflows, reduces verification time, and allows legislative bodies to manage a greater volume of petitions efficiently.

AUTHOR CONTRIBUTIONS

Elena Sánchez-Nielsen and Ismael Martín-Herrera: Conceptualization, methodology, software, validation, formal analysis, investigation, resources, and data curation. Elena Sánchez-Nielsen: Writing, visualization, supervision, project administration, and funding acquisition. All authors have read and agreed to the published version of the manuscript.

DATA AVAILABILITY STATEMENT

The FDD-S dataset is available at: <https://research-data.ull.es/datasets/zfb8mkxgrv/1> (doi: 10.17632/zfb8mkxgrv.1). The FDD-R dataset, which contains real personal data, is not publicly available.

CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

ABBREVIATIONS

The following abbreviations are used in this manuscript:

CIO	Chief Information Officer.
CNN	Convolutional Neural Network.
CV	Computer Vision.
ID	Personal Identifier.
FD	Form Document.
FFD-R	Real-World Form Document Dataset.
FFD-S	Synthetic Form Document Dataset.
GDPR	General Data Protection Regulation.
HSV	Handwritten Signature Verification.
NLP	Natural Language Processing.
OCR	Optical Character Recognition.
PLM	Pre-trained language model.
PoC	Proof of concept.
RNN	Recurrent Neural Network.
VVT	Verification, Validation, and Testing.
ULL	Universidad de La Laguna.

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